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**MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL
2026
ASSIGNMENT 1**

- Due date: 14 September (Monday), 22:00 pm.
- Write down your name, UID and the faculty to which you belong.
- Please submit your assignment on Moodle.
- You only have to turn in **1, 2, and 4**.
- Please include an **AI-use disclosure** at the end of your submission stating whether you used any AI tools, which tool(s) you used, exactly what parts they helped with (e.g., brainstorming, checking algebra, editing writing), and confirming that all final mathematical arguments are your own and that you can explain every step.

In this assignment, we construct the real numbers from the rational numbers using *Dedekind cuts*. You will define the basic operations, prove that they are well defined, and verify some of the ordered field axioms.

The assignment explains how to construct the set of real numbers via Dedekind cuts. We shall not check all axioms, but interested students should work out all the details to consolidate their understanding.

1. A *Dedekind cut* is a subset $\alpha \subsetneq \mathbb{Q}$ satisfying:

- (i) $\alpha \neq \emptyset$ and $\alpha \neq \mathbb{Q}$;
- (ii) if $x \in \alpha$ and $y < x$, then $y \in \alpha$;
- (iii) α has no maximum: for every $x \in \alpha$, there exists $z \in \alpha$ such that $x < z$.

Let $\mathbb{R}'$ denote the set of all Dedekind cuts.

- (a) Show that for each $r \in \mathbb{Q}$, the set

$$\hat{r} := \{q \in \mathbb{Q} : q < r\}$$

is a Dedekind cut.

- (b) Give an example of a Dedekind cut that is not equal to $\hat{r}$ for any $r \in \mathbb{Q}$.

2. **Addition.** For $\alpha, \beta \in \mathbb{R}'$, define

$$\alpha + \beta := \{q \in \mathbb{Q} : q = a + b \text{ for some } a \in \alpha, b \in \beta\}.$$

Prove that $\alpha + \beta$ is again a Dedekind cut. In other words, prove that addition is well defined on $\mathbb{R}'$.

3. (Optional) **Multiplication.** We say that a cut α is *nonnegative*, and write $\alpha \geq \hat{0}$, if

$$\hat{0} \subseteq \alpha.$$

For $\alpha, \beta \geq \widehat{0}$, define

$$\alpha\beta := \{q \in \mathbb{Q} : q < 0 \text{ or } (\exists a \in \alpha, a > 0)(\exists b \in \beta, b > 0) \text{ such that } q < ab\}.$$

Prove that if $\alpha, \beta \geq \widehat{0}$, then $\alpha\beta$ is a Dedekind cut. Thus multiplication is well defined for nonnegative cuts.

4. **Completeness.** Define an order relation $\leq$ on $\mathbb{R}'$ by

$$\alpha \leq \beta \quad \text{if and only if} \quad \alpha \subseteq \beta.$$

Let $\mathcal{S} \subseteq \mathbb{R}'$ be nonempty and bounded above. Define

$$u := \bigcup_{\alpha \in \mathcal{S}} \alpha.$$

You may use the fact that u is a Dedekind cut, without proving it. Show that u is the least upper bound of $\mathcal{S}$.

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P1 (a) (i) $\forall r \in \mathbb{Q}$, $r+1, r-1 \in \mathbb{Q}$, $r-1 < r < r+1$

$$\therefore r-1 \in \hat{r}, r+1 \notin \hat{r}$$

$$\text{i.e. } \hat{r} \neq \emptyset, \hat{r} \neq \mathbb{Q}$$

(ii) $\forall x \in \hat{r}$, we have $x < r$.

$$\therefore \forall y < x, y < r \text{ i.e. } y \in \hat{r}$$

(iii) Suppose z is maximum of $\hat{r}$.

$$\text{We have } \forall x \in \hat{r}, x \leq z.$$

Then by the denseness of $\mathbb{Q}$, $\exists z'$ s.t. $z < z' < r$.

which is a contradiction

$\therefore \hat{r}$ satisfies all three rules.

(b) An example is that: $S = \{q \in \mathbb{Q} : q^2 < 2\} \cup \emptyset$

Then we prove that $\nexists r \in \mathbb{Q}$ s.t. $S = \hat{r}$.

Suppose $S = \hat{r}$.

$$\textcircled{1} r^2 < 2. \text{ then } r \in S \text{ i.e. } S \neq \hat{r}$$

$$\textcircled{2} r^2 = 2. \text{ then } r \notin \mathbb{Q}, \text{ contradict.}$$

$$\textcircled{3} r^2 > 2. \text{ By the denseness of } \mathbb{Q}, \exists \frac{1}{n} < r^2 - 2.$$

$$\text{and } \exists \frac{1}{n'} < \frac{1}{(2r+1)n}$$

$$\therefore (r - \frac{1}{n'})^2 = r^2 - \frac{2r}{n'} + \frac{1}{(n')^2} > r^2 - \frac{2r+1}{n'} > r^2 - \frac{1}{n} > 2.$$

$$\text{i.e. } S = \widehat{(r - \frac{1}{n'})}, \text{ contradict.}$$

Q.E.D.

P2 Lemma: Any Dedekind cut is upper bounded.

prove: Suppose $\alpha \in \mathbb{R}'$ isn't upper bounded.

then $\forall q \in \mathbb{Q}$, if $q \notin \alpha$, then $\nexists p \in \alpha, p > q$

i.e. $\forall p \in \alpha, q > p$.

$\therefore q \in \alpha$ which means $\alpha = \mathbb{Q}$, contradict

(i) $\alpha, \beta \in \mathbb{R}' \therefore \alpha \neq \emptyset, \beta \neq \emptyset$. i.e. $\exists a \in \alpha, b \in \beta, a+b \in \alpha+\beta \neq \emptyset$

Let s be the sum of upper bounds of α and β in $\mathbb{Q}$

Obviously $\forall a \in \alpha, b \in \beta, s > a+b$.

$\therefore s \notin \alpha+\beta$. i.e. $\alpha+\beta \neq \mathbb{Q}$

(ii) $\forall x \in \alpha+\beta$, by the definition, $\exists a \in \alpha, b \in \beta, x = a+b$.

$\forall y < x, a - (x-y) < a$. $\therefore a - (x-y) \in \alpha$

i.e. $y = [a - (x-y)] + b \in \alpha+\beta$.

(iii) Suppose $\alpha+\beta$ has maximum m .

$\therefore \forall a' \in \alpha, b' \in \beta, m \geq a'+b'$ and $\exists a \in \alpha, b \in \beta, m = a+b$.

Then we prove that a must be the maximum of α

If $\exists a' > a, a' \in \alpha$, then $a'+b > a+b = m$ i.e. m isn't maximum of $\alpha+\beta$.

Which is contradict with $\alpha \in \mathbb{R}'$.

$\therefore \alpha+\beta$ hasn't maximum.

$\therefore \alpha+\beta \in \mathbb{R}' \forall \alpha, \beta \in \mathbb{R}'$

P3. (i) $\alpha, \beta \in \mathbb{R}' \therefore \alpha \neq \emptyset, \beta \neq \emptyset$. i.e. $\exists a \in \alpha, b \in \beta, ab \in \alpha\beta \neq \emptyset$

By the lemma in P2, α and β all have upper bound.

Let s be the product of upper bounds of α and β in $\mathbb{Q}$

$\therefore \alpha > 0, \beta > 0 \therefore$ The upper bounds must be positive.

Obviously $\forall a \in \alpha, b \in \beta, s > ab$

$\therefore s \notin \alpha\beta$ i.e. $\alpha\beta \neq \mathbb{Q}$

(ii) $\forall x \in \alpha\beta$, by the definition, $\exists a \in \alpha, b \in \beta, a > 0, b > 0, x = ab$

$\forall 0 < y < x, 0 < a \cdot \frac{y}{x} < a \therefore \frac{ay}{x} \in \alpha$ i.e. $y = \frac{ay}{x} \cdot b \in \alpha\beta$

$\forall y \leq 0 < x, y \in \alpha\beta$ by the definition

(iii) Suppose $\alpha\beta$ has maximum m .

$\therefore \forall a' \in \alpha, b' \in \beta, a' > 0, b' > 0, m \geq a'b'$ and $\exists a \in \alpha, b \in \beta, a > 0, b > 0, m = ab$

Then we prove that a must be the maximum of α

If $\exists a' > a, a' \in \alpha$, then $a'b > ab = m$ i.e. m isn't maximum of $\alpha\beta$

Which is contradict with $a \in \mathbb{R}'$.

$\therefore \alpha\beta$ hasn't maximum.

$\therefore \alpha\beta \in \mathbb{R}' \forall \alpha, \beta \in \mathbb{R}'$

P4. ① $\forall \beta \in S, \beta \in \bigcup_{\alpha \in S} \alpha = u$ i.e. $u \geq \beta$.

② Suppose $\exists u' < u$, u' is also an upper bound of S .

Then $\exists q \in \mathbb{Q}, q \in u$ but $q \notin u'$.

Otherwise $\forall x \in u, x \in u'$ i.e. $u \subseteq u'$ contradict.

Q.E.D.